

Лабораторная работа № 4.

Работа с программными пакетами

Работа с dnf и rpm

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Введение

Актуальность

Управление программным обеспечением является одной из ключевых задач системного администратора Linux. Пакетные менеджеры dnf и rpm позволяют устанавливать, обновлять, удалять и анализировать пакеты, а также управлять репозиториями.

Цель работы

Получить навыки работы с репозиториями и менеджерами пакетов dnf и rpm.

Задачи

1. Изучить, как и в каких файлах подключаются репозитории.
2. Изучить основные команды dnf.
3. Изучить расширенные возможности dnf.
4. Освоить работу с rpm.
5. Оформить отчёт с результатами и ответами на контрольные вопросы.

Объект и предмет исследования

Объект — программные пакеты и репозитории Linux. Предмет — инструменты управления пакетами dnf и rpm.

Методы исследования

Экспериментальные: выполнение команд в терминале, фиксация результатов скриншотами.

Теоретическая часть

Пакетный менеджер dnf используется в RPM-дистрибутивах (Fedora, RHEL, CentOS, AlmaLinux, Rocky Linux) для установки, удаления, обновления и анализа пакетов.

Основные команды dnf:

- dnf repolist — список подключённых репозиториях;
- dnf list — список пакетов;
- dnf search — поиск пакета;
- dnf info — информация о пакете;
- dnf install — установка пакета;
- dnf remove — удаление пакета;
- dnf history — история транзакций;
- dnf group list — список групп пакетов.

Утилита rpm работает с отдельными пакетами:

- rpm -Uhv — установить пакет;
- rpm -qi — информация о пакете;
- rpm -ql — список файлов пакета;
- rpm -qd — документация пакета;
- rpm -qc — конфигурационные файлы;
- rpm -q -scripts — скрипты установки/удаления;
- rpm -qf — определить пакет по файлу;
- rpm -e — удалить пакет.

Материалы и методы

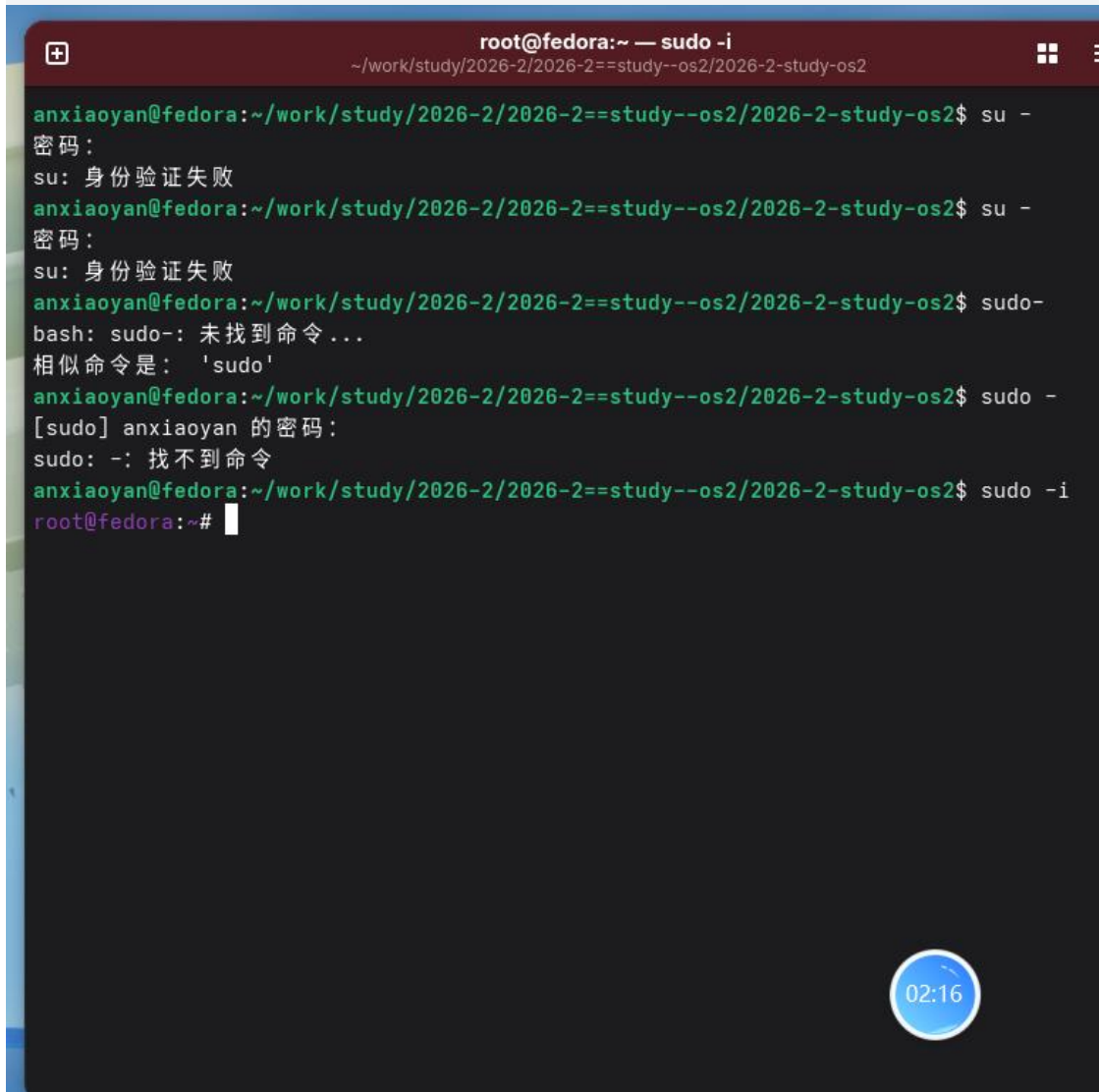
- оборудование: персональный компьютер;
- операционная система: Fedora 44;
- программное обеспечение: терминал, dnf5, rpm;
- метод сбора данных: выполнение команд и фиксация результатов скриншотами.

Выполнение лабораторной работы

Работа с репозиториями

Переход в режим суперпользователя:

```
sudo -i
```

A terminal window titled 'root@fedora:~ — sudo -i' with a path '~ /work/study/2026-2/2026-2==study--os2/2026-2-study-os2'. The user 'anxiaoyan' is at the prompt. They attempt to use 'su -' twice, but fail with 'su: 身份验证失败' (identity verification failed) and are prompted for a password. Then they try 'sudo -', which fails with 'bash: sudo-: 未找到命令...' (command not found) and suggests 'sudo'. Finally, they enter 'sudo -i', which succeeds after a password prompt, resulting in the root prompt 'root@fedora:~#'. A blue circular clock icon showing '02:16' is in the bottom right corner.

```
anxiaoyan@fedora:~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2$ su -
密码:
su: 身份验证失败
anxiaoyan@fedora:~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2$ su -
密码:
su: 身份验证失败
anxiaoyan@fedora:~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2$ sudo-
bash: sudo-: 未找到命令...
相似命令是: 'sudo'
anxiaoyan@fedora:~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2$ sudo -
[sudo] anxiaoyan 的密码:
sudo: -: 找不到命令
anxiaoyan@fedora:~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2$ sudo -i
root@fedora:~#
```

Переход в режим root

Просмотр каталога репозитория:

```
cd /etc/yum.repos.d
ls
cat fedora.repo
```

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-fedora-$releasever-$basearch
skip_if_unavailable=False

[fedora-debuginfo]
name=Fedora $releasever - $basearch - Debug
#baseurl=http://download.example/pub/fedora/linux/releases/$releasever/Everything/$basearch/debug/tree/
metalink=https://mirrors.fedoraproject.org/metalink?repo=fedora-debug-$releasever&arch=$basearch
enabled=0
metadata_expire=7d
repo_gpgcheck=0
type=rpm
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-fedora-$releasever-$basearch
skip_if_unavailable=False

[fedora-source]
name=Fedora $releasever - Source
#baseurl=http://download.example/pub/fedora/linux/releases/$releasever/Everything/source/tree/
metalink=https://mirrors.fedoraproject.org/metalink?repo=fedora-source-$releasever&arch=$basearch
enabled=0
metadata_expire=7d
repo_gpgcheck=0
type=rpm
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-fedora-$releasever-$basearch
skip_if_unavailable=False
root@fedora:/etc/yum.repos.d#
```

Содержимое каталога `/etc/yum.repos.d`

Список подключённых репозиториев:

```
dnf repolist
```

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

[fedora-source]
name=Fedora $releasever - Source
#baseurl=http://download.example/pub/fedora/linux/releases/$releasever/Everything/source/
metalink=https://mirrors.fedoraproject.org/metalink?repo=fedora-source-$releasever&basearch=
enabled=0
metadata_expire=7d
repo_gpgcheck=0
type=rpm
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-fedora-$releasever-$basearch
skip_if_unavailable=False
root@fedora:/etc/yum.repos.d# dnf repolist
repo id                                repo name
copr:copr.fedorainfracloud.org:phracek:PyCharm Copr repo for PyCharm owned by phracek
fedora                                  Fedora 44 - x86_64
fedora-cisco-openh264                  Fedora 44 openh264 (From Cisco) - x86_64
google-chrome                           google-chrome
rpmfusion-nonfree-nvidia-driver        RPM Fusion for Fedora 44 - Nonfree -
D
rpmfusion-nonfree-steam                 RPM Fusion for Fedora 44 - Nonfree -
updates                                Fedora 44 - x86_64 -
root@fedora:/etc/yum.repos.d#
```

Список репозиториев

Поиск пакетов по слову user:

```
dnf search user
```

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

updates Fedora 44 - x86_64 - Updates

root@fedora:/etc/yum.repos.d# dnf search user
仓库更新和加载中:
Fedora 44 - x86_64 - Updates          100% | 3.7 KiB/s | 9.4 KiB |
RPM Fusion for Fedora 44 - Nonfree - Steam 100% | 5.0 KiB/s | 8.4 KiB |
RPM Fusion for Fedora 44 - Nonfree - NVIDIA Driv 100% | 8.3 KiB/s | 8.8 KiB |
google-chrome                        100% | 1.6 KiB/s | 1.3 KiB |
Copr repo for PyCharm owned by phracek 100% | 2.7 KiB/s | 2.1 KiB |
仓库加载完成。
匹配字段: name, summary
atuin-all-users.noarch atuin init script for all users
bash-preexec-all-users.noarch bash-preexec init script for all users
cepces-user-enroll.noarch User Certificate Enrollment
flexmark-java-ext-gfm-users.noarch Flexmark extension for GitHub user syntax
ghc-xdg-userdirs.x86_64 Basic implementation of XDG user directories specifi
ghc-xdg-userdirs-devel.x86_64 Haskell xdg-userdirs library development files
ghc-xdg-userdirs-doc.noarch Haskell xdg-userdirs library documentation
ghc-xdg-userdirs-prof.x86_64 Haskell xdg-userdirs profiling library
gnome-user-docs.noarch GNOME User Documentation
gnome-user-share.x86_64 Gnome user file sharing
kf6-kuserfeedback.i686 Framework for collecting user feedback for apps via telemetr
surveys
kf6-kuserfeedback.x86_64 Framework for collecting user feedback for apps via
try and surveys
kf6-kuserfeedback-console.x86_64 Analytics and administration tool for UserFe
servers
kf6-kuserfeedback-devel.i686 Development files for kf6-kuserfeed
kf6-kuserfeedback-devel.x86_64 Development files for kf6-kuserfeed
kf6-kuserfeedback-doc.noarch Developer Documentation files for kf6-kuserfeedback
kf6-kuserfeedback-html.noarch Developer Documentation files for kf6-kuserfeedback
kf6-kuserfeedback-libs.x86_64 Libraries for kf6-kuserfeedback
```

Поиск пакетов по слову user

Установка и удаление пакета nmap

```
dnf info nmap
dnf install nmap
dnf install nmap\*
```

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

zypp-plugins.noarch    Plugins for libzypp users
zypp-tools.x86_64      Tools for libzypp users
root@fedora:/etc/yum.repos.d# dnf info nmap

仓库更新和加载中:
仓库加载完成。
可安装的软件包
名称      : nmap
时期      : 4
版本      : 7.92
发布      : 11.fc44
架构      : x86_64
下载大小  : 5.6 MiB
安装大小  : 23.8 MiB
来源      : nmap-7.92-11.fc44.src.rpm
仓库      : updates
概况      : Network exploration tool and security scanner
URL       : http://nmap.org/
协议      : LicenseRef-Nmap
描述      : Nmap is a utility for network exploration or security auditing. It supports ping scanning (determine which hosts are up), many port scanning techniques (determine what services the hosts are offering), and TCP/IP fingerprinting (remote host operating system identification). Nmap also offers flexible host and port specification, decoy scanning, determination of TCP sequence predictability characteristics, reverse-identd scanning, and more. In addition to the classic command-line nmap executable, the Nmap suite includes a file transfer, redirection, and debugging tool (netcat utility ncat), a utility for comparing scan results (ndiff), and a packet generation and response analysis tool (nping).
供应商    : Fedora Project

root@fedora:/etc/yum.repos.d# dnf install nmap
仓库更新和加载中:
仓库加载完成。
软件包      架构      版本      仓库
```

Информация о пакете nmap

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

概况      : Network exploration tool and security scanner
URL       : http://nmap.org/
协议      : LicenseRef-Nmap
描述      : Nmap is a utility for network exploration or security auditing. It supports
          : ping scanning (determine which hosts are up), many port scanning techniques
          : (determine what services the hosts are offering), and TCP/IP fingerprinting
          : (remote host operating system identification). Nmap also offers flexible target
          : and port specification, decoy scanning, determination of TCP sequence
          : predictability characteristics, reverse-identd scanning, and more. In addition
          : to the classic command-line nmap executable, the Nmap suite includes a flexible
          : data transfer, redirection, and debugging tool (netcat utility ncat), a utility
          : for comparing scan results (ndiff), and a packet generation and response
          : analysis tool (nping).
供应商    : Fedora Project

root@fedora:/etc/yum.repos.d# dnf install nmap
仓库更新和加载中:
仓库加载完成。
软件包      架构  版本      仓库      大小
正在升级:
  nmap-ncat    x86_64  4:7.92-11.fc44  updates    481.6 KiB
  替换 nmap-ncat x86_64  4:7.92-8.fc44   19278be6a81040f5b6c 481.5 KiB
正在安装:
  nmap         x86_64  4:7.92-11.fc44  updates    23.8 MiB

事务摘要:
正在安装:    1 个软件包
正在升级:    1 个软件包
正在替换:    1 个软件包

本次要处理的软件包总大小为 6 MiB。 还需下载 6 MiB。
此次操作完成之后, 24 MiB 额外的空间被使用 (安装 24 MiB, 删除 482 KiB)。
Is this ok [y/N]:
```

Установка пакета nmap

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

nmap-ncat                x86_64 4:7.92-11.fc44      updates                481.6 KiB
 替换 nmap-ncat          x86_64 4:7.92-8.fc44         19278be6a81040f5b6c 481.5 KiB
正在安装:
nmap                      x86_64 4:7.92-11.fc44      updates                23.8 MiB

事务摘要:
正在安装:    1 个软件包
正在升级:    1 个软件包
正在替换:    1 个软件包

本次要处理的软件包总大小为 6 MiB。 还需下载 6 MiB。
此次操作完成之后， 24 MiB 额外的空间被使用 (安装 24 MiB, 删除 482 KiB)。
Is this ok [y/N]: y
[1/2] nmap-ncat-4:7.92-11.fc44.x86_64          100% | 1.7 MiB/s | 230.5 KiB | 00m00s
[2/2] nmap-4:7.92-11.fc44.x86_64              100% | 6.9 MiB/s | 5.6 MiB | 00m01s
-----
[2/2] 总计                                     100% | 3.6 MiB/s | 5.8 MiB | 00m02s
执行事务
[1/5] 验证软件包文件                         100% | 33.0 B/s | 2.0 B | 00m00s
[2/5] 准备事务                               100% | 13.0 B/s | 3.0 B | 00m00s
[3/5] 升级 nmap-ncat-4:7.92-11.fc44.x86_64   100% | 3.7 MiB/s | 485.6 KiB | 00m00s
[4/5] 安装 nmap-4:7.92-11.fc44.x86_64        100% | 89.1 MiB/s | 24.0 MiB | 00m00s
[5/5] 删除 nmap-ncat-4:7.92-8.fc44.x86_64    100% | 9.0 B/s | 28.0 B | 00m03s
完成!
root@fedora:/etc/yum.repos.d# dnf install nmap/*
仓库更新和加载中:
仓库加载完成。
无法解析事务:
未找到匹配的参数: nmap/*
你可以尝试添加到命令行中:
--skip-unavailable 跳过不可用软件包
```

Установка всех пакетов nmap

Удаление пакетов:

```
dnf remove nmap
dnf remove nmap\*
```

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

[1/5] 验证软件包文件          100% | 33.0 B/s | 2.0 B | 00m00s
[2/5] 准备事务                100% | 13.0 B/s | 3.0 B | 00m00s
[3/5] 升级 nmap-ncat-4:7.92-11.fc44.x86_64 100% | 3.7 MiB/s | 485.6 KiB | 00m00s
[4/5] 安装 nmap-4:7.92-11.fc44.x86_64      100% | 89.1 MiB/s | 24.0 MiB | 00m00s
[5/5] 删除 nmap-ncat-4:7.92-8.fc44.x86_64 100% | 9.0 B/s | 28.0 B | 00m03s
完成!
root@fedora:/etc/yum.repos.d# dnf install nmap/*
仓库更新和加载中:
仓库加载完成。
无法解析事务:
未找到匹配的参数: nmap/*
你可以尝试添加到命令行中:
--skip-unavailable 跳过不可用软件包
root@fedora:/etc/yum.repos.d# dnf remove nmap
软件包          架构    版本          仓库          大小
正在删除:
nmap            x86_64   4:7.92-11.fc44 updates       23.8 MiB

事务摘要:
正在移除:    1 个软件包

此次操作完成之后, 24 MiB 空间被释放 (安装 0 B, 删除 24 MiB)。
Is this ok [y/N]: y
执行事务
[1/2] 准备事务                100% | 5.0 B/s | 1.0 B | 00m00s
[2/2] 删除 nmap-4:7.92-11.fc44.x86_64      100% | 568.0 B/s | 834.0 B | 00m01s
完成!
root@fedora:/etc/yum.repos.d# dnf remove nmap/*
没有与以下参数匹配的包可供删除: nmap/*

无需任何处理。
root@fedora:/etc/yum.repos.d#
```

Удаление nmap

Работа с группами пакетов

```
dnf5 group list
LANG=C dnf5 group info "rpm-development-tools"
dnf5 group install rpm-development-tools
```

```
root@fedora:/etc/yum/repos.d - sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

swaywm-extended      Sway Window Manager (supplemental packages)      no
system-tools         System Tools                                       no
text-internet        Text-based Internet                             no
vlc                  VideoLAN Client                                  no
window-managers      Window Managers                                  no
cosmic-desktop-apps  COSMIC Desktop Supplementary Applications         no
admin-tools          Administration Tools                             no
audio                Audio Production                                 no
authoring-and-publishing Authoring and Publishing                         no
budgie-desktop       Budgie                                           no
budgie-desktop-apps  Budgie Desktop Applications                     no
c-development        C Development Tools and Libraries               no
cloud-infrastructure Cloud Infrastructure                             no
cloud-management     Cloud Management Tools                          no
container-management Container Management                             yes
neuron-modelling-simulators Neuron Modelling Simulators                     no
compiz               Compiz                                           no
cosmic-desktop       COSMIC Desktop                                  no
3d-printing          3D Printing                                    no
d-development        D Development Tools and Libraries               no
desktop-accessibility Desktop accessibility                           yes
design-suite          Design Suite                                    no
development-tools    Development Tools                               no
domain-client        Domain Membership                               no
editors              Editors                                           no
education            Educational Software                             no
electronic-lab       Electronic Lab                                   no
engineering-and-scientific Engineering and Scientific                       no
freeipa-server       FreeIPA Server                                  no
games                Games and Entertainment                         no
headless-management  Headless Management                             no
root@fedora:/etc/yum/repos.d#
```

Список групп пакетов

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

window-managers      Window Managers      no
cosmic-desktop-apps  COSMIC Desktop Supplementary Applications  no
admin-tools          Administration Tools  no
audio                Audio Production     no
authoring-and-publishing  Authoring and Publishing  no
budgie-desktop        Budgie               no
budgie-desktop-apps    Budgie Desktop Applications  no
c-development         C Development Tools and Libraries  no
cloud-infrastructure   Cloud Infrastructure   no
cloud-management      Cloud Management Tools  no
container-management  Container Management   yes
neuron-modelling-simulators  Neuron Modelling Simulators  no
compiz               Compiz                no
cosmic-desktop        COSMIC Desktop        no
3d-printing           3D Printing           no
d-development         D Development Tools and Libraries  no
desktop-accessibility Desktop accessibility  yes
design-suite           Design Suite           no
development-tools     Development Tools     no
domain-client         Domain Membership     no
editors               Editors                no
education             Educational Software   no
electronic-lab        Electronic Lab         no
engineering-and-scientific  Engineering and Scientific  no
freeipa-server        FreeIPA Server         no
games                 Games and Entertainment  no
headless-management   Headless Management    no
root@fedora:/etc/yum.repos.d# LANG=C dnf5 group info "RPM
> Development Tools"
Updating and loading repositories:
Repositories loaded.
root@fedora:/etc/yum.repos.d#
```

Информация о группе

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2
--skip-unavailable 跳过不可用软件包
root@fedora:/etc/yum.repos.d# LANG=C dnf5 group list
Updating and loading repositories:
Repositories loaded.

```

ID	Name	Installed
network-server	Network Servers	no
kde-desktop	KDE	no
kde-education	KDE Educational applications	no
kde-media	KDE Multimedia support	no
kde-mobile	KDE Mobile	no
kde-mobile-apps	KDE Mobile Apps	no
kde-office	KDE Office	no
kde-pim	KDE PIM	no
kde-software-development	KDE Software Development	no
libreoffice	LibreOffice	yes
kf6-software-development	KDE Frameworks 6 Software Development	no
mate-applications	MATE Applications	no
milkymist	Milkymist	no
kde-apps	KDE Applications	no
office	Office/Productivity	no
python-classroom	Python Classroom	no
python-science	Python Science	no
robotics-suite	Robotics	no
rpm-development-tools	RPM Development Tools	no
security-lab	Security Lab	no
sound-and-video	Sound and Video	no
swaywm-extended	Sway Window Manager (supplemental packages)	no
system-tools	System Tools	no
text-internet	Text-based Internet	no
vlc	VideoLAN Client	no
window-managers	Window Managers	no
cosmic-desktop-apps	COSMIC Desktop Supplementary Applications	no
admin-tools	Administration Tools	no

Установка группы

Удаление группы:

```
dnf5 group remove rpm-development-tools
```

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

i3                                i3 window manager                                no
i3-extended                       i3 window manager (supplemental packages)        no
input-methods                     Input Methods                                     no
java                              Java                                                no
java-development                  Java Development                                  no
javaenterprise                    Java Application Server                          no
kde-apps                           KDE Applications                                  no
kde-desktop                        KDE                                                 no
engineering-and-scientific        Engineering and Scientific                        no
development-tools                 Development Tools                                 no
dial-up                           Dial-up Networking Support                       no
directory-server                  Directory Server                                  no
dns-server                        DNS Name Server                                  no
dogtag                             Dogtag Certificate System                        no
domain-client                     Domain Membership                                no
editors                           Editors                                             no
education                         Educational Software                             no
electronic-lab                     Electronic Lab                                     no
3d-printing                       3D Printing                                       no
enlightenment-desktop             Enlightenment                                     no
firefox                           Firefox Web Browser                               yes
fedora-packager                    Fedora Packager                                   no
fonts                             Fonts                                              yes
font-design                       Font design and packaging                        no
freeipa-server                     FreeIPA Server                                    no
ftp-server                         FTP Server                                        no
gnome-desktop                      GNOME                                              yes

root@fedora:/etc/yum.repos.d# LANG=C dnf5 group list | grep -i rpm
Updating and loading repositories:
Repositories loaded.
rpm-development-tools             RPM Development Tools                            no
root@fedora:/etc/yum.repos.d#
```

Удаление группы

История транзакций

```
dnf5 history list
dnf5 history undo 6
```

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

Updating and loading repositories:
Repositories loaded.
Id           : rpm-development-tools
Name          : RPM Development Tools
Description   : These tools include core development tools such rpmbuild.
Installed     : no
Order         :
Langonly      :
Uservisible   : yes
Repositories  : fedora, updates
Mandatory packages : redhat-rpm-config
                  : rpm-build
Default packages : koji
                  : mock
                  : rpmdevtools
Optional packages : punji
                  : rpmlint

root@fedora:/etc/yum.repos.d# dnf5 group install "RPM Development Tools"
仓库更新和加载中:
仓库加载完成。
无法解析事务:
未找到匹配的参数: RPM Development Tools
你可以尝试添加到命令行中:
--skip-unavailable 跳过不可用软件包
root@fedora:/etc/yum.repos.d# dnf5 remove rpm-build rpmdevtools rpmlint
没有与以下参数匹配的包可供删除: rpm-build
没有与以下参数匹配的包可供删除: rpmdevtools
没有与以下参数匹配的包可供删除: rpmlint

无需任何处理。
root@fedora:/etc/yum.repos.d#
```

История транзакций

```
root@fedora:/etc/yum.repos.d — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

4 dnf install -y kernel-devel kernel-headers gcc 2026-09-04 17:21:12 257
3 dnf5daemon-server 2026-06-19 19:05:20 7
2 dnf5 --config /kiwi_dnf5.config -y --disable-pl 2026-04-22 14:00:43 1721
1 dnf5 --config /builddir/result/image/build/imag 2026-04-22 13:58:52 278
root@fedora:/etc/yum.repos.d# dnf5 history undo 6
仓库更新和加载中：
仓库加载完成。
事务合并错误：「操作「Remove」 「nmap-4:7.92-11.fc44.x86_64」因该时间点不存在而无法合并 -
跳过。」

软件包          架构    版本          仓库          大小
正在降级：
nmap-ncat       x86_64    4:7.92-8.fc44  fedora        481.5 KiB
  替换 nmap-ncat  x86_64    4:7.92-11.fc44  updates      481.6 KiB

事务摘要：
正在替换：    1 个软件包
正在降级：    1 个软件包

本次要处理的软件包总大小为 231 KiB。 还需下载 231 KiB。
此次操作完成之后，8 B 空间被释放（安装 482 KiB，删除 482 KiB）。
Is this ok [y/N]: y
[1/1] nmap-ncat-4:7.92-8.fc44.x86_64 100% | 1.8 MiB/s | 231.0 KiB | 00m00s
-----
[1/1] 总计 100% | 210.9 KiB/s | 231.0 KiB | 00m01s
执行事务
[1/4] 验证软件包文件 100% | 47.0 B/s | 1.0 B | 00m00s
[2/4] 准备事务 100% | 9.0 B/s | 2.0 B | 00m00s
[3/4] 降级 nmap-ncat-4:7.92-8.fc44.x86_64 100% | 4.2 MiB/s | 485.6 KiB | 00m00s
[4/4] 删除 nmap-ncat-4:7.92-11.fc44.x86_64 100% | 26.0 B/s | 28.0 B | 00m01s
完成！
root@fedora:/etc/yum.repos.d#
```

Отмена транзакции

Работа с rpm на примере lynx

```
dnf5 install lynx --downloadonly
find /var/cache/libdnf5/ -name lynx\*
```

```
fedora-updates-testing.repo
google-chrome.repo
rpmfusion-nonfree-nvidia-driver.repo
rpmfusion-nonfree-steam.repo
root@fedora:/etc/yum.repos.d# cd /var/cache/libdnf5/fedora-cff72538bc9825a4/packages
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# ls
libklvanc-1.6.0-4.fc44.x86_64.rpm  lynx-2.9.2-5.fc44.x86_64.rpm
llhttp-9.3.1-1.fc44.x86_64.rpm    sso-mib-libs-0.8.0-1.fc44.x86_64.rpm
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -Uhv
rpm: 未给出要安装的软件包
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -Uhv
rpm: 未给出要安装的软件包
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -Uhv lynx-2.9.2-5.fc
44.x86_64.rpm
  校验中 ..... ##### [100%]
  准备中 ... ##### [100%]
  正在升级/安装 ...
    1:lynx-2.9.2-5.fc44 ##### [100%]
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# which lynx
/usr/sbin/lynx
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -qf $(which lynx)
lynx-2.9.2-5.fc44.x86_64
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -qi lynx
Name       : lynx
Version    : 2.9.2
Release    : 5.fc44
Architecture: x86_64
Install Date: 2026年09月25日 星期五 12时55分20秒
Group      : Unspecified
Size       : 6357914
License    : GPL-2.0-only
Signature  :
```

Скачивание пакета lynx

```
/usr/share/locale/sv/LC_MESSAGES/lynx.mo
/usr/share/locale/tr/LC_MESSAGES/lynx.mo
/usr/share/locale/uk/LC_MESSAGES/lynx.mo
/usr/share/locale/vi/LC_MESSAGES/lynx.mo
/usr/share/locale/zh_CN/LC_MESSAGES/lynx.mo
/usr/share/locale/zh_TW/LC_MESSAGES/lynx.mo
/usr/share/man/man1/lynx.1.gz
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -qd lynx
/usr/share/doc/lynx/README
/usr/share/doc/lynx/docs/CHANGES2.3
/usr/share/doc/lynx/docs/CHANGES2.4
/usr/share/doc/lynx/docs/CHANGES2.5
/usr/share/doc/lynx/docs/CHANGES2.6
/usr/share/doc/lynx/docs/CHANGES2.7
/usr/share/doc/lynx/docs/CHANGES2.8
/usr/share/doc/lynx/docs/CRAWL.announce
/usr/share/doc/lynx/docs/FM.announce
/usr/share/doc/lynx/docs/IBMPC-charsets.announce
/usr/share/doc/lynx/docs/README.TRST
/usr/share/doc/lynx/docs/README.chartrans
/usr/share/doc/lynx/docs/README.cookies
/usr/share/doc/lynx/docs/README.defines
/usr/share/doc/lynx/docs/README.metrics
/usr/share/doc/lynx/docs/README.options
/usr/share/doc/lynx/docs/README.release
/usr/share/doc/lynx/docs/README.rootcerts
/usr/share/doc/lynx/docs/README.ssl
/usr/share/doc/lynx/docs/README.sslcerts
/usr/share/doc/lynx/docs/SOCKETSHR.announce
/usr/share/doc/lynx/docs/TCPWARE.announce
/usr/share/doc/lynx/docs/VMSWAIS.announce
/usr/share/doc/lynx/docs/djgpp.key
```

Поиск скачанного пакета

Установка через rpm:

```
rpm -Uhv lynx-2.9.2-5.fc44.x86_64.rpm
```

Запросы к пакету:

```
which lynx
rpm -qf $(which lynx)
rpm -qi lynx
rpm -ql lynx
rpm -qd lynx
rpm -qc lynx
rpm -q --scripts lynx
```

```
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

/usr/share/doc/lynx/test/bad-html.html
/usr/share/doc/lynx/test/c1.html
/usr/share/doc/lynx/test/circle.html
/usr/share/doc/lynx/test/cp-1252.html
/usr/share/doc/lynx/test/cp-1252a.html
/usr/share/doc/lynx/test/idna-tr46.html
/usr/share/doc/lynx/test/image.jpg
/usr/share/doc/lynx/test/iso-8859-1.html
/usr/share/doc/lynx/test/iso-8859-1a.html
/usr/share/doc/lynx/test/iso-8859-2.html
/usr/share/doc/lynx/test/iso-8859-2a.html
/usr/share/doc/lynx/test/koi8-r.html
/usr/share/doc/lynx/test/nobody
/usr/share/doc/lynx/test/quickbrown.html
/usr/share/doc/lynx/test/raw8bit.html
/usr/share/doc/lynx/test/sgml.html
/usr/share/doc/lynx/test/spaces.html
/usr/share/doc/lynx/test/special_urls.html
/usr/share/doc/lynx/test/square.html
/usr/share/doc/lynx/test/tabtest.html
/usr/share/doc/lynx/test/tags.html
/usr/share/doc/lynx/test/test-styles.html
/usr/share/doc/lynx/test/triangle.html
/usr/share/doc/lynx/test/unicode.html
/usr/share/doc/lynx/test/utf-8-demo.html
/usr/share/man/man1/lynx.1.gz
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -qc lynx
/etc/lynx-site.cfg
/etc/lynx.cfg
/etc/lynx.lss
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -q --scripts lynx
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages#
```

Запросы к пакету lynx

Удаление:

```
rpm -e lynx
```

Работа с rpm на примере dnsmasq

```
dnf5 install dnsmasq
which dnsmasq
rpm -qf $(which dnsmasq)
rpm -qi dnsmasq
rpm -ql dnsmasq
rpm -qd dnsmasq
rpm -qc dnsmasq
rpm -q --scripts dnsmasq
```

```
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages x anxiaoyan@fedora:~/work/study/2026-2/2026-2==study-os2

仓库更新和加载中：
仓库加载完成。
软件包「dnsmasq-2.92-4.fc44.x86_64」已安装。

无需任何处理。
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# which dnsmasq
/usr/sbin/dnsmasq
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -qf $(which dnsmasq)
dnsmasq-2.92-4.fc44.x86_64
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -qi dnsmasq
Name       : dnsmasq
Version    : 2.92
Release    : 4.fc44
Architecture: x86_64
Install Date: 2026年 04月 22日 星期三 17时 04分 28秒
Group      : Unspecified
Size       : 840959
License    : GPL-2.0-only OR GPL-3.0-only
Signature  :
            RSA/SHA256, 2026年 01月 16日 星期五 18时 35分 59秒, Key ID dbfcf71c6d9f90a6
Source RPM : dnsmasq-2.92-4.fc44.src.rpm
Build Date : 2026年 01月 16日 星期五 18时 30分 29秒
Build Host : buildvm-x86-17.rdu3.fedoraproject.org
Packager   : Fedora Project
Vendor     : Fedora Project
VCS        : git:git://thekelleys.org.uk/dnsmasq.git
URL        : http://www.thekelleys.org.uk/dnsmasq/
Bug URL    : https://bugz.fedoraproject.org/dnsmasq
Summary    : A lightweight DHCP/caching DNS server
Description:
Dnsmasq is lightweight, easy to configure DNS forwarder and DHCP server.
```

Установка и запросы к dnsmasq

При удалении через `rpm -e` возникла ошибка зависимостей. Использован `dnf5 remove dnsmasq`:

```
dnf5 remove dnsmasq
```

```
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages — sudo -i
~/work/study/2026-2/2026-2==study--os2/2026-2-study-os2

root@fedora:/var/cache/libdnf5/fedora-cff72538bc x anxiaoyan@fedora:~/work/study/2026-2/2026-2==stu

>>> Warning: The unit file, source configuration file or drop-ins of virtnwfilterd-ro.socket changed on disk. Run 'systemctl daemon-reload' to reload units.
>>> Warning: The unit file, source configuration file or drop-ins of virtnwfilterd-admin.socket changed on disk. Run 'systemctl daemon-reload' to reload units.
>>>
[94/97] 删除 libvirt-daemon-driver-nwfilter-0: 100% | 184.0 B/s | 19.0 B | 00m00s
>>> 运行 %preun, 包: libvirt-daemon-driver-secret-0:12.0.0-3.fc44.x86_64
>>> %preun 运行完成, 包: libvirt-daemon-driver-secret-0:12.0.0-3.fc44.x86_64
>>> 包输出:
>>> Removed '/etc/systemd/system/sockets.target.wants/virtsecret-admin.socket'.
>>> Removed '/etc/systemd/system/sockets.target.wants/virtsecret-ro.socket'.
>>> Removed '/etc/systemd/system/sockets.target.wants/virtsecret.socket'.
>>> Disabling 'virtsecret.service', but its triggering units are still active:
>>> virtsecret.socket, virtsecret-ro.socket, virtsecret-admin.socket
>>> Warning: The unit file, source configuration file or drop-ins of virtsecret.service changed on disk. Run 'systemctl daemon-reload' to reload units.
>>> Warning: The unit file, source configuration file or drop-ins of virtsecret.socket changed on disk. Run 'systemctl daemon-reload' to reload units.
>>> Warning: The unit file, source configuration file or drop-ins of virtsecret-ro.socket changed on disk. Run 'systemctl daemon-reload' to reload units.
>>> Warning: The unit file, source configuration file or drop-ins of virtsecret-admin.socket changed on disk. Run 'systemctl daemon-reload' to reload units.
>>>
[95/97] 删除 libvirt-daemon-driver-secret-0:12 100% | 531.0 B/s | 17.0 B | 00m00s
[96/97] 删除 libvirt-ssh-proxy-0:12.0.0-3.fc44 100% | 312.0 B/s | 5.0 B | 00m00s
[97/97] 删除 libportal-gtk3-0:0.9.1-4.fc44.x86 100% | 2.0 B/s | 6.0 B | 00m02s
完成!
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages# rpm -q dnsmasq
未安装软件包 dnsmasq
root@fedora:/var/cache/libdnf5/fedora-cff72538bc9825a4/packages#
```

Удаление dnsmasq

Результаты

1. Изучены файлы репозитория.
2. Получен список репозитория.
3. Установлен и удалён пакет nmap.
4. Установлена и удалена группа rpm-development-tools.
5. Выполнена отмена транзакции dnf5.
6. Установлен, проанализирован и удалён lynx.
7. Установлен, проанализирован и удалён dnsmasq.

Обсуждение результатов

Все команды выполнены успешно. dnf5 автоматически разрешает зависимости и ведёт историю. rpm требует ручного разрешения зависимостей.

Заключение

- Цель работы достигнута.
- Изучены репозитории, группы пакетов, история транзакций.
- Освоены установка, анализ и удаление пакетов.

Приложение. Ответы на контрольные вопросы

1. `dnf provides "*/useradd"` — поиск пакета с файлом `useradd`.
2. `dnf5 group list`, `dnf5 group info "security-tools"` — список групп и информация.
3. `rpm -ivh package.rpm` — установка скачанного пакета.
4. `rpm -q --scripts package_name` — просмотр скриптов.
5. `rpm -qd package_name` — документация пакета.
6. `rpm -qf /path/to/file` — пакет по файлу.